IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

COMMUNICATION NETWORKS

Friday, 13 December 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : C. Ling

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Information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,
$1/\mu$ = Average length of packet [bit/packet]
C_i = Transmission speed link i [bits/s]
μC_i = Service rate (link i) [packet/ s]
λ_i = Arrival rate (link i) [packet/ s]

2. Optimal Routing Problem (ORP)

$Min\ D(F)$ with respect to $F = \{ F_i \}$

where, $D(F) = \sum_{i=1}^{L} \frac{F_i}{C_i - F_i}$

and,

C_i = Capacity of link l_i.
F_i = Flow carried by link l_i.

3. Jain's fairness index

$$\boldsymbol{f(x_1, x_2, \dots, x_n) = \frac{(\sum_{i=1}^{n} x_i)^2}{n \sum_{i=1}^{n} x_i^2}}$$

Note: each user $\boldsymbol{i}$ receive $\boldsymbol{x_i}$

3. Velocity of propagation and propagation delay.

Velocity of unguided transmission through air or space of an electromagnetic wave is $\boldsymbol{2.99x10^8}$ m/s.

Assume velocity of guided transmission to be approximately $\boldsymbol{66}$% of the unguided wave propagation.

The Questions

Q1.

a) For the go back N ARQ mechanism, if a frame is in error, it will generate K retransmissions. For this mechanism the average numbers of retransmissions N_r can be expressed as:

$$N_r = \frac{1 - P + KP}{1 - P}$$

i) Explain the meaning of P. **[2]**

ii) Using as a reference the parameter $\boldsymbol{a}$ given by:

$$\boldsymbol{a} = \frac{\boldsymbol{Rd}}{\boldsymbol{LV}}$$

where,
$\boldsymbol{R}$ = Data rate of the channel.
$\boldsymbol{d}$ = Length of the network.
$\boldsymbol{L}$ = Average frame length.
$\boldsymbol{V}$ = Velocity of guided media.

Assuming that $\boldsymbol{a}$ has been normalised:
estimate the value of K when the window size is $W \geq 2a + 1$ and when the window size is $W < 2a + 1$. **[3]**

b)

For a multiservice DS network, the behaviour of an edge router is different from the behaviour of a core router.

i) Discuss in your own words the functionalities that an edge router implements. **[3]**

ii) Identify which QoS guarantee-network mechanism(s) are being observed by the edge router functionalities. **[3]**

iii) Discuss in your own words the functionalities that a core router implements. **[3]**

iv) Identify which QoS guarantee-network mechanism(s) are being observed by the core router functionalities. **[3]**

v) Briefly compare Intserv vs Diffserv approach to support QoS in terms of I) coordination of service differentiation, II) scope of service differentiation and III) scalability. **[3]**

Q2.

a)

i) For the network shown in Fig. 2.1. solve the optimal routing problem. Assume all the links have capacity 10 Kbps and the traffic requirements are as per figure. **[3]**

ii) Derive the minimum capacity of link $C(1,3)$ such that all the traffic between nodes 1 and 3 is carried only by link $C(1,3)$. **[5]**

b) The total energy consumption of a K node wireless network is given by the following expression:

$$\text{Total Power} = \sum_{k=1}^{k} (E_t + E_r)\, B$$

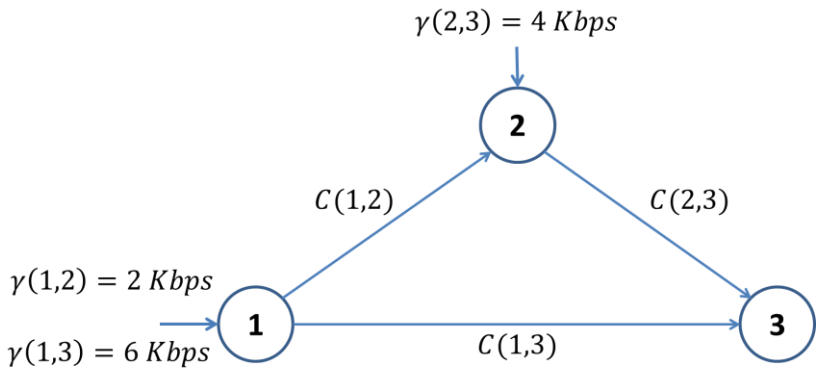
For the transmitter: $E_t = e_t + e_d r^3$.
For the receiver: $E_r = e_r$.

Where the values of the parameters are:
$e_t = e_r = 5 \cdot 10^{-10}$ Jules/bit.
$e_d = 100 \cdot 10^{-12}$ Jules/bit/m^2.
$B = 2$ MBit/s.
$r =$ distance in m between communicating nodes.

i) For the network shown in **Fig. 2.2.a.** and considering only the energy consumption associated to parameter e_d: identify the lowest energy consumption path of a communication between nodes **A** and **E**. **[5]**

ii) Place now node **C** in the middle of the straight line of sight between nodes **A** and **E** (**Fig.2.2.b.**). For this new topology, identify the lowest energy consumption path of a communication between nodes **A** and **E**. Discuss your findings. **[4]**

iii) Suggest the best possible position of nodes **B** and **D** that will minimise the energy consumption path of a communication between nodes **A** and **E** (no need for calculations). Clearly argue your proposed solution. **[3]**



$$C(1,2) = C(1,3) = C(2,3) = 10\ Kbps$$

Fig. 2.1.: Network Q2a).

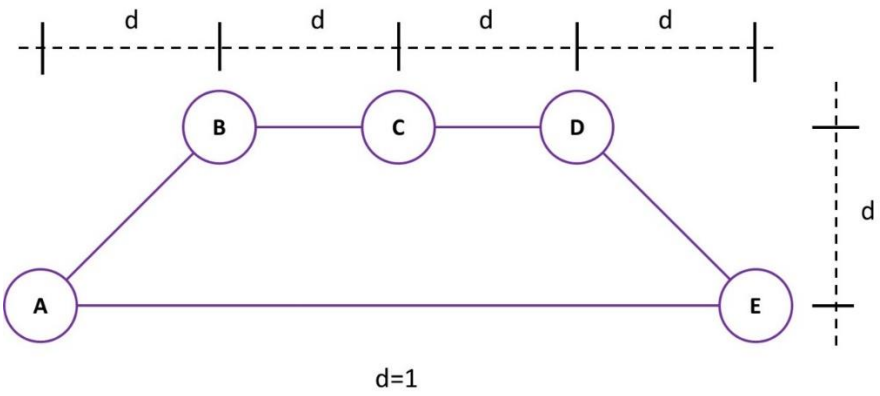


Figure 2.2.a: Wireless network, where $d = 1$.

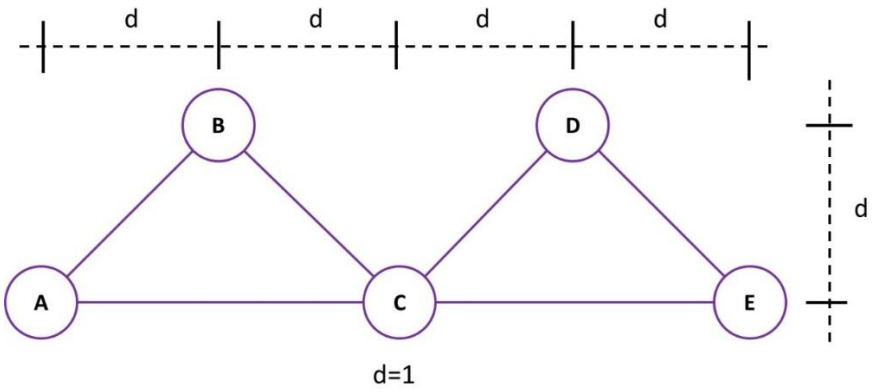


Figure 2.2.b: Wireless network, new position node C, where $d = 1$.

Q3

a)

i) Describe and discuss what is the distinctive characteristic of a statistical shortest path problem. **[2]**

ii) For a given network all its links are characterised by the mean and variance of independent Gaussian distributions.

I) You are required to solve the shortest path of the cost function $\mu_p + \sigma_p^2$ (i.e. mean + variance).
II) You are required to solve the shortest path of the cost function $\mu_p + \sigma_p$ (i.e. mean + standard deviation).

Discuss which type of algorithms (stochastic or deterministic) will you use in case I) and in case II) and why? **[4]**

b)

i) Describe and discuss a packet scheduling mechanism known to you that is max-min fair when arriving packets are all the same size. **[2]**

ii) What is the value of the Jain index when k out of n users receive all available resources? And what will be the value of most unfairness allocation? **[3]**

For the remainder of this question, you need to allocate bandwidth among three users:

User 1: represents an elastic traffic.

$$U_1(x) = 1 - e^{-4x}$$

User 2: represents an adaptive real-time traffic.

$$U_2(x) = 1 - \frac{1}{e^{20(x-0.4)} + 1}$$

User 3: represents a real-time traffic.

$$U_3(x) = u(x - 0.6)$$

Assume the operational domain $x \in (0,1)$ and operational range $U_i(x) \in (0,1)$.

iii) Plot the three normalised utility functions. **[2]**

iv) Identify in the same plot a utility max-min allocation (e.g. $U_i(x) = 0.5$). Discuss the impact on all the users. **[3]**

v) Identify in the same plot the bandwidth max-min allocation if you are allowed an allocation up to $x = 0.5$ and $x = 0.7$. Discuss the impact on all user of the allocated bandwidth. **[4]**

Q4.

a)

i) Briefly describe under which condition(s) a standard TCP protocol will have to reduce its congestion window size. Explain the rationale of the decision. **[4]**

ii) In the same context of a standard TCP protocol discuss the importance of properly designing the time-out parameters and how will you implement an estimation of it. **[4]**

b)

For the packet network in **Fig. 4.1** all external arrivals are assumed Poisson streams with mean arrival rate γ_{ij} packets per second. The traffic matrix T is given:

$$T = \begin{bmatrix} 0 & 0 & \gamma_{13} & \gamma_{14} \\ \gamma_{21} & 0 & \gamma_{23} & 0 \\ \gamma_{31} & \gamma_{32} & 0 & 0 \\ 0 & 0 & \gamma_{43} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 3 & 3 \\ 2 & 0 & 1 & 0 \\ 2 & 1 & 0 & 0 \\ 0 & 0 & 3 & 0 \end{bmatrix}$$

All links have capacity: $C(i,j)$ = 5 packets per second.

i) Identify the bottleneck link(s). **[3]**

ii) Derive the mean number of outstanding packets in the network. **[4]**

iii) If you are required to take out of service (remove) one link to carry out preventive maintenance: discuss the reasons behind your choice and numerically evaluate how the original operational point of the networks has been affected. **[5]**

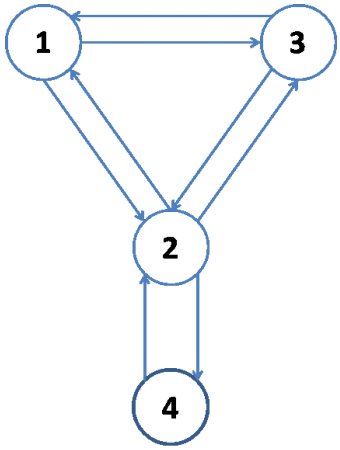


Figure 4.1.